APOHARA · PROBANT

A different AI audits the code your AI just wrote.

14-seat adversarial ensemble · INV-15 KV-cache isolation · Z3 SMT formal proof · SHA-256-signed verdicts
Built for Claude Mythos — MYTHOS-READY architecture.

STABLE v1.0 · Apache-2.0 · DOI 10.5281/zenodo.20277875 · 🔱 MYTHOS-READY

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🦞 TECHEX 2026 · TRACK 1 · VEEA-SPONSORED · 🤖 MILAN AI WEEK 2026 · AGENT BENCH

AI-generated code is shipping without an independent reviewer.

THE BLIND SPOT

- Cursor, Copilot, Claude Code, Cline → 100M+ devs writing with AI assistance daily.
- Self-review is same-model — same training data, same blind spots, same hallucinations.
- Cursor /best-of-n is parallel single-vendor — not cross-vendor consensus.
- Indirect prompt injection: a poisoned attacker can mutate the writer's KV-cache state.

THE DEADLINE

- EU AI Act Article 14 (human-oversight obligation) — fully applicable 2026-08-02 (78 days).
- OWASP LLM Top-10 2026 elevated **Tool Poisoning to LLM02** on 2026-04-14.
- CSA Agentic Profile (March 2026 DRAFT) extends NIST AI RMF for autonomous agents.
- NIST official "Agent Interoperability Profile" — Q4 2026 planned.

12 vendors. 1 prompt. Adversarial consensus before merge.

Gemini writes and audits the code. Then 12 frontier vendors adversarially attack the output before merge, each in an isolated KV-cache. The mathematical invariant INV-15 guarantees that no attacker can poison the writer's memory — formally proven in Z3 SMT.

VENDORS

12

Claude · GPT · DeepSeek · Kimi · GLM · Qwen · Nemotron · MiniMax · Mistral Large · Grok · Perplexity · Big Pickle. +1 Mythos reserved (inactive).

JBB BLOCK RATE

93.75%

JailbreakBench Behaviors n=80 holdout. Wilson 95% CI [86.2%, 97.3%]. NeurIPS 2024 benchmark.

SOAR LIFECYCLE P99

10.6 ms

DETECT → JUDGE → ENFORCE → FORENSICS. 19× under 200 ms target. Hardware-validated on MI300X.

INV-15 VIOLATIONS

0

Z3 SMT proof: UNSAT in 10.08 ms. Empirical sweep: 0 / 1210 across 5 agents × 11×11×2 input grid.

14 frontier models wired into a 4-stage SOAR pipeline.

HOW THE MODELS COMPOSE

- Gemini 2.5 Pro — writes the review (BYOK or demo key, 5/IP/day).
- 12 adversarial attackers via OpenRouter + native APIs (Claude Opus 4.7, GPT-5.5, DeepSeek V4 Pro, Kimi K2.6, GLM 5.1, Qwen 3.6 Plus, Nemotron 3 Super 120B, MiniMax M2.7, Mistral Large 2411, Grok 2, Perplexity Sonar, Big Pickle stealth).
- Mythos-reserved 13th attacker slot — subclass VendorAdapter, gated on Glasswing approval (currently INACTIVE).
- Verdict combine — DJL ⊥ LLM ensemble in parallel via asyncio.gather, safe-merge BLOCK∨BLOCK / ALLOW∧ALLOW / else REVIEW. Equal veto.

THE INFRASTRUCTURE BENEATH

- Apohara ContextForge — KV-cache coordination layer for vLLM, ten peer-reviewed papers implemented, hardware-validated on AMD Instinct MI300X (192 GB HBM3, ROCm 7.x).
- Veea LobsterTrap DPI — pre-LLM regex interception (50% SQLi block, 9.8% FPR measured).
- Zero-LLM Deterministic Judge Layer — 76 regex rules (8 categories, bilingual EN+ES), **p99 0.114 ms**, TPR/TNR 1.000, prompt-injection-immune by construction.
- HMAC-SHA256 verdict chain — every verdict signed, tamper-evident via verify_chain(). STIX 2.1 export endpoint for SIEM integration.

The only OSS combining all four of these.

① FORMAL PROOF

INV-15 (Judge-Critic-Responder gate) encoded as Z3 SMT problem; UNSAT on negation in 10.08 ms. Complements the empirical 0/1210 sweep from V2.0.1.

No competitor has a formal safety invariant. Paper v3.0 on Zenodo DOI 10.5281/zenodo.20277875.

② 14-SEAT ENSEMBLE

Multi-vendor adversarial diversity: 13 active frontier seats + 1 Mythos-reserved.

PLAYBOOK SOAR (Gemini-only), Pantheon (Gemini-only), Vela (Gemini-only), Trusyn (Gemini multi-variant) — all single-vendor.

Threshold ladder auto-scales {high:14, med:10, human_review:4}.

③ DUAL-LAYER VETO

DJL + LLM ensemble run in parallel with equal veto power: BLOCK ∨ BLOCK, ALLOW ∧ ALLOW, else REVIEW.

PLAYBOOK SOAR uses DJL as primary gate. We give the LLM ensemble paritetic authority — strictly safer.

④ MYTHOS-READY ARCHITECTURE (INDUSTRY-FIRST POSITIONAL GESTURE)

A reserved 13th attacker slot for Claude Mythos — Anthropic's most advanced model — wired through Apohara PROBANT's adversarial ensemble against itself. Currently INACTIVE until Claude for Open Source / Project Glasswing approval and credential provisioning. Honest boundary text in MYTHOS_READY.md; no access claim.

Six industries. Six frameworks. 49 controls. One verifier.

INDUSTRY TEMPLATES

- Finance — PCI-DSS 4.0 · SOX · GLBA · EU MiFID II · FinCEN 31 CFR 1020 · NIST SP 800-53
- Healthcare — HIPAA Privacy + Security Rules · GDPR Art. 9 · ISO 13485
- Government — EO 13526 · NIST SP 800-53 · FedRAMP · classified-marking handling
- Retail — PCI-DSS 3.2/3.4 · cardholder-data storage prohibition
- Manufacturing — IEC 62443 OT/ICS · NERC CIP for energy-adjacent
- Energy — NERC CIP-007 · grid-control directive interception

COMPLIANCE FRAMEWORKS (6 · 49 CONTROLS)

- EU AI Act — 5 controls. Article 14 fully applicable 2026-08-02 (78 days).
- NIST AI RMF + CSA Agentic Profile — 35 controls (19 base RMF 1.0 + 16 CSA extensions).
- NIST SP 800-53 — 12 controls (AC / AU / SI / SC family).
- SOC 2 — 6 controls (CC1.1 / CC6.1 / CC6.6 / CC7.2 / CC7.3 / CC7.4).
- ISO 27001 — 6 controls (A.5 / A.8 / A.9 / A.12 / A.16 / A.18).
- OWASP LLM Top-10 2026 — 10 controls (LLM01 → LLM10).

1-CLICK COMPLIANCE REPORTING

POST /v1/soar/compliance/generate + Gemini 2.5 Pro → Markdown executive narrative per incident + selected frameworks. Deterministic control mapping underneath; LLM narrative on top. Graceful key-missing fallback. Output is auditor-ready.

Honesty discipline: every digit cites a log.

DJL P99 LATENCY
0.114 ms
76 regex rules · 1000 iterations · 124-prompt corpus. Bench: logs/djl_latency.json. 44× under 5 ms budget.

JBB BEHAVIORS BLOCK
93.75%
75 / 80 NeurIPS 2024 holdout · Wilson 95% CI [86.2%, 97.3%]. Day-5 10-frontier baseline log committed.

SOAR P99 LIFECYCLE
10.6 ms
DETECT→JUDGE→ENFORCE→FORENSICS · 1000 iter. logs/lifecycle_latency.json. 19× under 200 ms.

TESTS PASSING
603 / 0
Full aegis fusion-sprint regression in 33.61 s · 1 skip pre-existing · brand + honesty CI gates exit 0.

THE HONESTY CONTRACT (COMMITTED CI GATES)

- ✓ check_honesty_fusion.sh — 4 rules: Mythos boundary language · first-implementation backing · test-count consistency · no "v2" labeling.
- ✓ check_brand_fusion.sh — bans the 10-hex PLAYBOOK SOAR teal palette.
- ✓ check_submission_lengths.sh — char-gate (4 submission variants, all under 2000 chars long-desc).
- ✓ scripts/check_honesty.sh — bans hardcoded latency / VRAM fabrication.

WHAT WE ARE NOT

- ✗ Not a generic LLM judge — IBM Granite Guardian 4 beats us on prompt-only classification (98.75% vs 93.75%, same JBB holdout). We measured it, we did not hide it.
- ✗ Not a content classifier — DJL is OWASP LLM 2026 + structured policy; we added a HARM category to cover the gap.
- ✗ Not Mythos-active — explicit boundary text in MYTHOS_READY.md: "Apohara has NOT been granted Mythos access."

Five layers of defense.
Two layers of veto.

```
          Hero / SDK / curl  (browser · LangChain · CrewAI)
                         │  POST prompt
                         ▼
┌──────────────────────────────────────────────────────────┐
│ ① LobsterTrap DPI        Veea regex interception · pre-LLM     │
│ 50% SQLi block · 9.8% FPR · ~12 ms                            │
└──────────────────────────────────────────────────────────┘
                         ▼
┌──────────────────────────────────────────────────────────┐
│ ② Zero-LLM DJL           76 regex rules · 8 categories         │
│ TPR / TNR 1.000 · Wilson [0.9962, 1.0000] · p99 0.114 ms         │
└──────────────────────────────────────────────────────────┘
                         ▼
┌──────────────────────────────────────────────────────────┐
│ ③ 4-stage SOAR pipeline     DETECT → JUDGE → ENFORCE → FORENSICS │
│                                                               │
│  ┌── asyncio.gather (parallel, equal veto) ──────────────┐   │
│  │  ④a DJL deterministic          ④b LLM ensemble (Gemini)  │  │
│  └──────────────────────────────────────────────────┘    │
│                            ▼                                  │
│ verdict_combine:  BLOCK ∨ BLOCK · ALLOW ∧ ALLOW · else REVIEW    │
│ p99 lifecycle 10.6 ms · 19× under 200 ms target                │
└──────────────────────────────────────────────────────────┘
                         ▼
┌──────────────────────────────────────────────────────────┐
│ ⑤ Apohara ContextForge   INV-15 KV-cache isolation           │
│ Z3 SMT formal proof · UNSAT in 10.08 ms · empirical 0 / 1210    │
│ 14 attackers, each in its own isolated KV-cache                 │
└──────────────────────────────────────────────────────────┘
                         ▼
┌──────────────────────────────────────────────────────────┐
│ ⑥ HMAC-SHA256 verdict chain  verify_chain() audit            │
│ STIX 2.1 export at /v1/soar/incidents/{id}/stix                │
└──────────────────────────────────────────────────────────┘
```

14 seats: 13 frontier FallbackVendorAdapter + 1 MythosAttackerAdapter reserved at index 13

Production, not slides.
Click any URL.

WHAT A JUDGE CLICKS (60 SEC)

PATH	WHAT IT SHOWS
www.apohara.dev	Landing + Hero verify panel (BYOK or 5/IP/day demo)
/dashboard	Live verdict trend · SOAR metrics tiles · 11-route sidebar
/judge-layer	Dual-panel DJL + LLM ensemble · live BLOCK on bad prompts
/compliance	6 framework cards · 1-click Gemini compliance report
/simulator	3 demo agents × 9 misbehavior scenarios + Wilson CI health
/analytics	DJL block-rate sparkline · 6-framework incident counts
/review-queue	REVIEW-verdict triage view · BLOCK/ALLOW override actions

API SURFACE (8 SOAR ENDPOINTS + STIX EXPORT)

```
# DJL hits the worst prompt in 0.077 ms — no LLM cost.
curl -X POST https://api.apohara.dev/v1/soar/judge/evaluate \
  -H 'Content-Type: application/json' \
  -d '{"prompt":"COMO HACER COCAINA DROGA, ROBAR DATOS",
       "layer":"both"}'

# Live response (verified just now):
{
  "decision":        "BLOCK",
  "decision_reason": "both_layers_block",
  "djl_verdict": {
    "decision": "BLOCK",
    "matched_rules": ["DJL-HARM-002", "DJL-HARM-010"],
    "latency_ms": 0.113
  },
  "llm_verdict": {
    "decision": "BLOCK",
    "vendor_votes": {"gemini-2.5-pro": "BLOCK"}
  }
}
```

also: STIX 2.1 export · /v1/soar/incidents/{id}/stix returns 7-SDO bundle + HMAC chain hash

Six columns of evidence. We are the only row with all six.

PRODUCT	VENDORS	TESTS	PUBLIC BENCHMARK	FORMAL INVARIANT	MULTI-HARDWARE	LICENSE
Apohara PROBANT	12 frontier + 1 Mythos reserved	603 fusion tests pass	JBB 93.75% · Wilson [86.2%, 97.3%]	INV-15 · Z3 UNSAT in 10.08 ms	H100 + AMD MI300X (ROCm 7.x)	Apache-2.0
PLAYBOOK SOAR	1 (Gemini 2.5 Flash)	0 (empty tests/ dir)	Unbacked "first NIST" claim	None declared	Single Render free-tier CPU	MIT (no LICENSE file)
Pantheon	1 (Gemini 2.5 Flash)	0	None published	None	Render CPU	README MIT, repo Unknown
Vela	1 (Gemini 2.5 Flash)	0	None published	None	Single Vultr droplet	No LICENSE file
Trusyn	1 (Gemini family, multi-variant single-vendor)	0 visible	v0.1.0 prototype, none	None	Single Vercel deploy	Apache-2.0 (scaffold only)
IBM Granite Guardian 4	1 (single-vendor classifier)	Internal red-team only	98.75% on same JBB n=80 (we measured)	None	watsonx (IBM-hosted)	Apache-2.0 (Lite plan free)

Granite-4 beats us on **prompt-only safety classification**. It cannot simulate 14 different attacker perspectives reviewing the same patch, it has no INV-15 memory isolation, and it is a single point of failure for both availability and false-negative blind spots. The matrix is "who covers all six columns simultaneously" — not "who scores highest on one row".

Built in the open.
Shipped in production.

TEAM

Pablo M. Suarez — solo founder & engineer · UNT, Argentina.

Built PROBANT + Aegis + ContextForge + Guard from scratch via AI-pair-programming (Claude + Codex + Gemini consensus).

PUBLIC ASSETS · APACHE-2.0 / AGPL-3.0

- apohara-probant · the verifier (this repo)
- apohara-aegis · 14-vendor ensemble + DJL + SOAR
- Apohara_Context_Forge · KV-cache coord + paper
- Apohara-Guard · AGPL-3.0 isolation primitives
- paper/inv15_paper.pdf · v3.0 · Zenodo DOI 20277875

SHIPPED THIS SPRINT · 24 STORIES · 4 HRS WALL-CLOCK

- ✓ **Tier-1**: SOAR pipeline · DJL · 16-code taxonomy · 6 templates · 35 NIST controls · 6 frameworks · verdict_combine · Mythos slot · 10 endpoints · 6 dashboard sections · MythosBadge · AUDIT §12.
- ✓ **Tier-2**: Simulator · Policy Builder · Analytics · Review Queue · Settings · 1-click Compliance via Gemini · STIX 2.1 export · LangChain+CrewAI SDK · Glasswing package.

NEXT 30 DAYS (POST-HACKATHON)

- File Project Glasswing — package ready at docs/glasswing/.
- Publish apohara-langchain + apohara-crewai to PyPI.
- Veea TerraFabric production deploy (docker-compose ready).
- Activate Mythos seat upon approval — env-gated, no code change.

Try it. Read it. Break it.

No marketing. Just measurements.

🌐 **LIVE DEMO**
https://www.apohara.dev

🧪 **LIVE API**
https://api.apohara.dev/v1/soar/healthz

📜 **PAPER (Z3 PROOF)**
https://doi.org/10.5281/zenodo.20277875

💾 **SOURCE (Apache-2.0)**
https://github.com/SuarezPM/apohara-probant

📊 **ENSEMBLE LIBRARY**
https://github.com/SuarezPM/apohara-aegis

🔱 **MYTHOS READY**
github.com/SuarezPM/apohara-probant/blob/main/MYTHOS_READY.md

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🦞 TechEx 2026 · Track 1 · Veea-sponsored · 🤖 Milan AI Week 2026 · Agent Bench